
(a)
Serial inference cost
SPGFusion
516.3 ms
4 detector
forwards
130.0 ms
706.3 ms
0
200
400
600
800
Serial latency (ms)
(b)
Candidate cost profile
1600
1200
800
400
0
Extra peak memory (MiB)
SPGFusion
PIAFusion
Detector
30
50
100
300
500
Latency (ms, log scale)
(c)
Deployment decision boundary
Validation calibration stable
Calibration unstable
Evaluator + domain stable
Always CDF
fixed evaluator + stable validation
Visible / high shield
recalibrate before switching
Evaluator or domain changes
CS-ECF
revalidate on held-out groups
Visible / high shield
unstable calibration
Latency-sensitive deployment
reject four-candidate search unless pruning or early termination is measured
After fine-tuning, revalidate before deployment.